

1. Basic Counting

(AP) / (MP) – the two engines

(**AP**) Split into *disjoint* cases, add counts.
(**MP**) Build in *independent ordered steps*, multiply counts.

Intuition: AP = OR (union of disjoint bins), MP = AND (a chain of independent choices).

(CP) / (DP) – run them backward

- (**CP**) $|U \setminus A| = |U| - |A|$: count the complement when it's easier.
- (**DP**) If a procedure with m outcomes hits every object exactly r times, the number of objects is m/r .

DP is MP in reverse: overcounting by a constant factor r means dividing by r .

Permutations

$$P_r^n = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}$$

r ordered picks without replacement from n items.

Circular permutations (unnamed seats)

$$Q_r^n = \frac{P_r^n}{r} = \frac{n!}{r(n-r)!}, \quad Q_n^n = (n-1)!$$

Divide by r rotations (DP). “Glue a block”: k tied objects $\rightarrow k!$ internal orders.

- n couples, alternating: $(n-1)!n!$
- n couples, each next to spouse: $2^n(n-1)!$ ($n \geq 2$; careful at $n = 1$: answer is 1, not 2)

Combinations

$$\binom{n}{r} = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!} = \binom{n}{n-r}$$

$$\sum_{r=0}^n \binom{n}{r} = 2^n$$

Choose r unordered from n : build via ordered pick, then DP by $r!$ (kill the order).

Pascal: $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$ (split on whether elt. n is chosen)

Double-counting identities (count 2 ways):

- $r\binom{n}{r} = n\binom{n-1}{r-1}$
- $(n-r+1)\binom{n}{r-1} = r\binom{n}{r}$
- $(n-r)\binom{n}{r} = n\binom{n-1}{r}$
- $\binom{n}{m}\binom{m}{r} = \binom{n}{r}\binom{n-r}{m-r}$

Trick: count pairs (a subset, a flagged element inside/outside it) two different orders.

No two consecutive: # of r -subsets of $[n]$ with no 2 consecutive $= \binom{n-r+1}{r}$. Shift $a_i \mapsto a_i - (i-1)$ to kill the gaps – turns it into an ordinary r -subset of a smaller set.

Pairings of $2n$ people into n unordered pairs:

$$\frac{(2n)!}{2^n n!} = (2n-1)!!$$

Line up, cut into pairs: divide out 2^n (order within pair) and $n!$ (order of pairs).

PIE

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A_1 \cup \cdots \cup A_q| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots$$

Each element in k sets gets counted $\binom{k}{1} - \binom{k}{2} + \cdots = 1$ time, net.

Euler’s φ : $n = p_1^{m_1} \cdots p_k^{m_k} \Rightarrow \varphi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right)$ PIE on “divisible by p_i ”; only distinct primes matter, exponents don’t.

(IP) / (BP)

- Injection $A \rightarrow B \Rightarrow |A| \leq |B|$.
- Bijection $A \rightarrow B \Rightarrow |A| = |B|$.

Workhorse: count A by matching it to a set B you already know how to count. Check injective (read object back off image) + surjective (every image hit).

Lattice paths $X \rightarrow Y$ (m E-steps, n N-steps): $\binom{m+n}{n}$ (word in E’s/N’s).

Subsets: $|\mathcal{P}(X)| = 2^n$ via characteristic vectors in $\{0,1\}^n$.

2. The Twelffold Way

r objects into n boxes; objects/boxes each distinct or identical; no restriction / injective (each box ≤ 1) / surjective (each box ≥ 1).

	No restr.	Inject.	Surject.
Obj D, Box D	n^r	P_r^n	$n!\left\{ \begin{smallmatrix} r \\ n \end{smallmatrix} \right\}$
Obj D, Box I	$\sum_{k=0}^n \left\{ \begin{smallmatrix} r \\ k \end{smallmatrix} \right\}$	$[r \leq n]$	$\left\{ \begin{smallmatrix} r \\ n \end{smallmatrix} \right\}$
Obj I, Box D	$\binom{r+n-1}{r}$	$\binom{n}{r}$	$\binom{r-1}{n-1}$
Obj I, Box I	$\sum_{k=0}^n p(r,k)$	$[r \leq n]$	$p(r,n)$

$[r \leq n]$ means 1 if $r \leq n$, else 0. D=distinct, I=identical.

Stars and bars (Obj I, Box D)

$$\binom{r+n-1}{r}$$

$\#$ arrangements of r identical stars and $(n-1)$ bars. Same count answers: nonneg. int. solutions of $x_1 + \cdots + x_n = r$;

multisets of size r from n types (H_r^n). Surjective version (no box empty): bars go in $r-1$ *internal* gaps only:

$$\binom{r-1}{n-1}$$

Multisets & permutations

- Permutations of $\{r_1 \cdot a_1, \dots, r_n \cdot a_n\}$, $r = \sum r_i$: $\frac{r!}{r_1! \cdots r_n!}$
- r -perms. of $\{\infty \cdot a_1, \dots, \infty \cdot a_n\}$: n^r

Stirling numbers, 2nd kind $\left\{ \begin{smallmatrix} r \\ n \end{smallmatrix} \right\}$
partitions of r distinct elts. into n nonempty unlabelled blocks.

$$\left\{ \begin{smallmatrix} r \\ n \end{smallmatrix} \right\} = n \left\{ \begin{smallmatrix} r-1 \\ n \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} r-1 \\ n-1 \end{smallmatrix} \right\}, \quad \left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$$

Split on element r : alone in new block $\left(\left\{ \begin{smallmatrix} r-1 \\ n-1 \end{smallmatrix} \right\}\right)$, or joins one of n existing blocks $\left(n \left\{ \begin{smallmatrix} r-1 \\ n \end{smallmatrix} \right\}\right)$. **Explicit (via PIE on surjections):**

$$\left\{ \begin{smallmatrix} r \\ n \end{smallmatrix} \right\} = \frac{1}{n!} \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} j^r$$

$n! \left\{ \begin{smallmatrix} r \\ n \end{smallmatrix} \right\} = \#$ surjections $[r] \rightarrow [n]$ (label the blocks). **Falling factorial:** $x^{\underline{r}} = x(x-1)\cdots(x-r+1)$

$$x^r = \sum_{k=0}^r \left\{ \begin{smallmatrix} r \\ k \end{smallmatrix} \right\} x^{\underline{k}}$$

Count functions $[r] \rightarrow [m]$ two ways: directly m^r , or by image size k . **Bell number** $B_r = \sum_k \left\{ \begin{smallmatrix} r \\ k \end{smallmatrix} \right\} = \#$ all partitions of an r -set.

Stirling numbers, 1st kind (unsigned) $\left[\begin{smallmatrix} r \\ n \end{smallmatrix} \right]$
permutations of r elts. with exactly n disjoint cycles.

$$\left[\begin{smallmatrix} r \\ n \end{smallmatrix} \right] = (r-1) \left[\begin{smallmatrix} r-1 \\ n \end{smallmatrix} \right] + \left[\begin{smallmatrix} r-1 \\ n-1 \end{smallmatrix} \right]$$

Insert elt. r : start new cycle alone $\left(\left[\begin{smallmatrix} r-1 \\ n-1 \end{smallmatrix} \right]\right)$, or slot into 1 of $r-1$ arc-positions of an existing cycle $\left((r-1) \left[\begin{smallmatrix} r-1 \\ n \end{smallmatrix} \right]\right)$. **Signed:** $s(r,n) = (-1)^{r-n} \left[\begin{smallmatrix} r \\ n \end{smallmatrix} \right]$, and

$$x^{\underline{r}} = \sum_{k=0}^r s(r,k) x^k$$

2nd-kind matrix $\left[\left\{ \begin{smallmatrix} r \\ k \end{smallmatrix} \right\}\right]$ and 1st-kind matrix $[s(r,k)]$ are inverse to each other.

Integer partitions $p(r,n)$

ways to write r as an unordered sum of n positive parts.

$$p(r,n) = p(r-1,n-1) + p(r-n,n)$$

Split on smallest part: $= 1$ (delete it, $p(r-1,n-1)$) or ≥ 2 (strip 1 from every part, $p(r-n,n)$). Ferrers diagram: rows = parts, left-aligned, decreasing. Conjugate (transpose) swaps #rows $\leftrightarrow$ largest part $\Rightarrow$ #partitions of r into n parts = #partitions of r with largest part n . No closed form for $p(r,n)$ or $p(r) = \sum_k p(r,k)$ exists (see Ch.3 for generating-function attack).

3. Binomial Thm & Generating Functions

Binomial & multinomial theorems

$$(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

Expand unsimplified: each of n factors contributes an x or y ; a string with r y ’s is chosen in $\binom{n}{r}$ ways. Plug in constants to get identities:

- $x = y = 1$: $\sum_r \binom{n}{r} = 2^n$
- $x = 1, y = -1$: $\sum_r (-1)^r \binom{n}{r} = 0$ ($n \geq 1$)
- $\Rightarrow$ even-index sum = odd-index sum $= 2^{n-1}$
- differentiate & set $x = 1$: $\sum_r r \binom{n}{r} = n 2^{n-1}$

Multinomial:

$$(x_1 + \cdots + x_m)^n = \sum \binom{n}{n_1, \dots, n_m} x_1^{n_1} \cdots x_m^{n_m}$$

$$\binom{n}{n_1, \dots, n_m} = \frac{n!}{n_1! \cdots n_m!}$$

Distribute n distinct positions among m letters; $n_1 + \cdots + n_m = n$.

Generating functions – setup

$F(x) = \sum_n a_n x^n$ encodes sequence (a_n) ; $[x^r]F(x) = a_r$. Cauchy product: $FG \leftrightarrow c_r = \sum_{k=0}^r a_k b_{r-k}$.

Product interpretation: if $x_i \in A^{(i)}$ has factor $F^{(i)}(x) = \sum_{k \in A^{(i)}} x^k$, then

$$\begin{aligned} [x^r] F^{(1)}(x) \cdots F^{(m)}(x) \\ = \# \{x_1 + \cdots + x_m = r : x_i \in A^{(i)}\} \end{aligned}$$

Weighted version: use x^{ik} for a variable with coefficient i (e.g. coin values, dice weights).

Encoding dictionary:

constraint	factor
none	$1/(1-x)$
$x_i \geq a$	$x^a/(1-x)$
$x_i \leq b$	$(1-x^{b+1})/(1-x)$
$a \leq x_i \leq b$	$(x^a - x^{b+1})/(1-x)$
x_i even	$1/(1-x^2)$
x_i mult. of k	$1/(1-x^k)$

Extracting coefficients (the toolkit)

- **Cumulative sum:** $\frac{1}{1-x} F(x) \leftrightarrow c_r = a_0 + \cdots + a_r$
- **Stars and bars:** $\frac{1}{(1-x)^n} = \sum_{r \geq 0} \binom{r+n-1}{n-1} x^r$
- **Shifting:** $[x^r] x^m F(x) = [x^{r-m}] F(x)$
- **Partial fractions, distinct roots:** $\frac{P(x)}{\prod (1-r_i x)} = \sum \frac{A_i}{1-r_i x}$, and $\frac{1}{1-rx} = \sum r^n x^n$.

- Repeated root, mult. m :**

$$\sum_n \binom{n+j-1}{j-1} r^n x^n$$

$$\frac{1}{(1-rx)^j} =$$

Recipe: (1) write down $F(x)$ from the dictionary, (2) simplify/factor, (3) apply the toolkit.

Integer partitions via GF

$$\sum_{r \geq 0} p(r) x^r = \prod_{k \geq 1} \frac{1}{1-x^k}$$

Factor k tracks $m_k = \#$ parts equal to k ; multiplicities are the exponents. **Euler:** $\#$ partitions into distinct parts = $\#$ partitions into odd parts. *Proof:* $D(x) = \prod (1+x^k) = \prod \frac{1-x^{2k}}{1-x^k}$ telescopes to $\prod \frac{1}{1-x^{2m-1}} = O(x)$.

Exponential generating functions (EGF)

$$\sum_{r \geq 0} a_r \frac{x^r}{r!}$$

OGF $\rightarrow$ selection/distribution problems (order irrelevant).
EGF $\rightarrow$ arrangement problems (order matters).

Product rule: EGF of “mix several kinds of objects in an arrangement” = product of each kind’s EGF (binomial convolution: coeff. $\sum_i \binom{r}{i} a_i b_{r-i}$).

Dictionary (type may appear):

restriction	EGF factor
any $\#$ times	e^x
at most n times	$1 + \frac{x}{1!} + \cdots + \frac{x^n}{n!}$
at least once	$e^x - 1$
even $\#$ times	$(e^x + e^{-x})/2$
odd $\#$ times	$(e^x - e^{-x})/2$

Read off coefficients using $e^{nx} = \sum_r n^r x^r / r!$. This is PIE done by algebra.

4. Recurrence Relations

From counting problem to recurrence

Find the size parameter n ; look at the “last piece” (last tile, largest disk, last element); split into disjoint cases by (AP); get initial conditions from small cases. **Tiling $1 \times n$ with $1 \times 1, 1 \times 2$:** $a_{n+2} = a_{n+1} + a_n$ (Fibonacci).

Tower of Hanoi: $a_n = 2a_{n-1} + 1$, $a_1 = 1 \Rightarrow a_n = 2^n - 1$.

Linear homogeneous recurrences

$c_0 a_n + c_1 a_{n-1} + \cdots + c_r a_{n-r} = 0$. Try $a_n = x^n \Rightarrow$

Characteristic eqn: $c_0 x^r + c_1 x^{r-1} + \cdots + c_r = 0$

Main Thm.

- **(I) distinct roots $\alpha_1, \dots, \alpha_r$:** $a_n = A_1 \alpha_1^n + \cdots + A_r \alpha_r^n$
- **(II) root α_i w/ multiplicity m_i :** contributes $(A_{i1} + A_{i2}n + \cdots + A_{im_i} n^{m_i-1}) \alpha_i^n$

Then plug in initial conditions to solve for the A ’s (linear system). Complex roots come in conjugate pairs $re^{\pm i\theta}$; combine to real form $2r^n \cos(n\theta + \phi)$, or just keep as $A(re^{i\theta})^n + \bar{A}(re^{-i\theta})^n$.

Fibonacci ($a_0 = a_1 = 1$): roots $\frac{1 \pm \sqrt{5}}{2}$, Binet’s formula $a_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{n+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{n+1} \right]$.

General (non-homogeneous) linear recurrences

$c_0 a_n + \cdots + c_r a_{n-r} = f(n)$.

General solution = (homog. solution $h(n)$) + (one particular solution $g_0(n)$)

Difference of any two solutions solves the homogeneous equation – so this always works, like ODEs. **Trial forms for $g_0(n)$** (base a ; if a is a root of mult. m , multiply trial by n^m):

$f(n)$	trial $g_0(n)$
$P_k(n)$ (poly., $a=1$)	$Q_k(n)$
$c a^n$	$C a^n$
$P_k(n) a^n$	$Q_k(n) a^n$
$f_1 + f_2$	$g_{0,1} + g_{0,2}$ (superposition)

Substitute trial into recurrence, match coefficients.

Generating function method (handles everything at once)

Steps: (1) mult. recurrence by x^n , sum over valid $n \Rightarrow$ linear eqn. for $A(x)$; (2) solve for $A(x)$ (rational function); (3) partial fractions $\rightarrow$ read off a_n .

Bookkeeping:
$$\sum_{n \geq k} a_{n-k} x^n = x^k A(x),$$

$$\sum_{n \geq k} a_n x^n = A(x) - \sum_{i < k} a_i x^i$$

Common OGFs: $a_n=1 \rightarrow \frac{1}{1-x}$; $a_n=n \rightarrow \frac{x}{(1-x)^2}$;

$a_n=r^n \rightarrow \frac{1}{1-rx}$. Denominator of $A(x)$ = characteristic polynomial read backwards ($c_0 + c_1 x + \cdots + c_r x^r$); a repeated root of mult. m shows up as $(1-\alpha x)^m$ in the denominator, matching Main Thm. part (II) automatically. Non-homog. $f(n)$ just adds to the numerator – no separate particular-solution guess needed.

Partial fractions recap:

$$\frac{P(x)}{(1-r_1x)(1-r_2x)} = \frac{A}{1-r_1x} + \frac{B}{1-r_2x}$$

$$A = \frac{P(1/r_1)}{1-r_2/r_1}, \quad B = \frac{P(1/r_2)}{1-r_1/r_2}$$

Systems of recurrences

When one sequence’s recurrence needs an auxiliary sequence, either (a) eliminate the auxiliary variable algebraically to get one

higher-order recurrence, or (b) write both as generating-function equations and solve the linear system for $A(x)$ (usually cleaner). Both sequences share the same denominator (hence the same characteristic roots / growth rate); only the numerator (initial data) differs.

Quick sanity checklist

- AP: are the cases really *disjoint* and *exhaustive*?
- MP: are the choices really *independent* of earlier ones?
- DP/circular: does every group really have the *same* size r ? (couples $n=1$ trap!)
- GF: did you get the encoding factor right per variable *before* multiplying?
- Recurrence: does the trial form need an extra n^m because the base is a root?